Semantic Joint Monocular Remote Sensing Image Digital Surface Model Reconstruction Based on Feature Multiplexing and Inpainting

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***Abstract*— Digital surface model (DSM) presents height information of the Earth's surface and plays an important role in many remote sensing (RS) applications. Since the conventional acquisition of DSM is laborious and expensive, DSM reconstruction from monocular RS images has attracted extensive research in recent years, which is an ill-posed problem and thus rather challenging. Related works have achieved great accomplishments in this regard; however, they still face some limitations in training robustness, accuracy, and efficiency. To address the issues, a semantic joint monocular RS image DSM regression framework is proposed in this article, whose salient points include that: 1) semantic segmentation is integrated into the DSM regression task so that a shared backbone can extract complementary features from each objective to improve the performance of the individual task. Meanwhile, based on the consistency of the two training objectives, a two-stage joint loss function is introduced to improve the convergence and robustness of model training; 2) an encoding–decoding backbone is designed based on feature multiplexing, which simultaneously achieves multiscale feature fusion and information decoupling, thereby greatly reducing model parameters and improving efficiency while ensuring feature extraction effect; and 3) an iterative upsampling approach is introduced to transform the full-scale spatial features into large receptive-field and locally discriminative dynamic kernels, which are used to inpaint coarse-grained features while decoding, thus enhancing regression accuracy. Finally, experiments demonstrate the effectiveness of the proposal. It is easy to train and achieves superior or comparable accuracy compared with state-of-the-art related works while improving the efficiency by a large margin.**

***Index Terms*— Digital surface model (DSM) reconstruction, feature inpainting upsampling (FIU), feature multiplexing, joint multitask, monocular remote sensing (RS) image.**

I. Introduction

DIGITAL surface model (DSM) is a widely used data product in photogrammetry and remote sensing (RS) that presents the height information of natural and artificial ground objects on the earth's surface [1], which plays an important role in a variety of applications such as semantic labeling [2], [3], scene understanding [4], 3-D modeling [5], change detection [6]–[8], and so on. Commonly, the DSM is directly obtained with dedicated deployments such as a light detection and ranging (LiDAR) laser scanner or an interferometric synthetic-aperture radar (InSAR) [9], or generated from stereo pairs of consecutive-view optical aerial images or high-resolution satellite images via sophisticated photogrammetric image techniques like structure from motion (SfM) [10], [11]. Therefore, on a routine basis, the DSM acquisition is considerably laborious and expensive, and it is theoretically not feasible to generate DSM from historical RS images because methods such as SfM are not available when overlapping stereo pairs cannot be acquired.

The observation and perception technologies of the earth's surface have undergone rapid development in recent years, and a large number of optical RS images serve many fields of social production [12]. Compared with conventional methods, it will be quite economical, convenient, and flexible to reconstruct DSM from monocular optical RS images, that is, to generate height maps from single-view multispectral aerial or satellite images (or their specific band combination). In fact, this kind of thinking has received more and more attention, and scholars have conducted a lot of research in this regard [1], [8], [13]–[27]. Monocular RS image DSM reconstruction is a mathematically ill-posed problem as there is an inherent ambiguity in mapping an intensity or color measurement into a height value [13], thus being rather challenging. Most existing methods implement the height estimation in a semantic-segmentation fashion because there is a strong correlation between the two tasks. Specifically, semantic segmentation (SS) in computer vision (CV) is the task of linking each pixel in the image to a semantic category label. Currently, with the rise of deep learning technology, SS based on convolutional neural networks (CNNs) has been extensively studied and achieved state-of-the-art (SOTA) results. SS CNN usually adopts an encoding–decoding structure. The encoder gradually downsamples the input image to generate feature maps that contain high-level and coarse-grained semantic information. The decoder gradually upsamples the encoded feature maps to the decoded feature maps, which contain low-level and fine-grained information on the basis of high-level semantic information, and are finally used for

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pixel-wise classification. The height information of the ground objects in the real world is also associated with their semantics. So, similar to the SS task, the height value of each pixel in the monocular RS image can also be estimated by an encoding–decoding CNN, except that the former is a classification problem while the latter is a regression problem. In particular, Mou and Zhu [13] proposed to transform the input monocular RS image to high-level feature representations with a convolutional subnetwork and generate the height map from the extracted features with a deconvolutional subnetwork. Meanwhile, skip connections between the subnetworks were adopted to preserve fine edge details. Amini Amirkolaee and Arefi [1] exploited an encoding–decoding CNN to generate the preliminary DSM of the single-view high-resolution satellite image and presented a filtering approach based on morphological operators to extract the nonground pixels from the estimated height map. Li *et al.* [15] proposed to divide height values into space-increasing intervals and transform the regression problem into a classification problem. A CNN was used to predict the height-discrete patches which are then spliced into a seamless height map. Pellegrin and Martinez-Carranza [14] judiciously selected and processed small patches from existing depth estimation datasets to train CNNs with basic encoding–decoding architectures, and the models are proved to be feasible in monocular aerial image height prediction. Gao and Shen [20] proposed a semisupervised StyHighNet for height estimation of single-view aerial images, which transfers multisource images to a unified style distribution for height map prediction. Paoletti *et al.* [19] presented to generate high-resolution DSMs from RS images based on the generative adversarial network. A pair of corresponding RS images and DSM are introduced into the encoder branches for mapping into the shared-latent representation, and from which the predictions are recovered by the decoder branches.

The above methods have achieved great accomplishments. However, they still face some limitations of robustness, accuracy, and efficiency in engineering applications.

1) If an encoding–decoding CNN is directly used to make pixel-wise dense height prediction, this article asserts that the model training will be unrobust and the regression loss will be difficult to converge. Undoubtedly, regression is a more complicated problem than classification, since discrete values are more tolerant of prediction errors than continuous values, while regression is more sensitive to outliers. For the DSM regression task, the prediction of each pixel of the input image will be used as a positive sample to participate in the calculation of training loss. When the number of the regression values is so large, outliers abound, and prediction errors are hard to tradeoff and reconcile. Therefore, the training is prone to gradient explosion, and the regression loss will oscillate wildly.
2) Mainstream backbones utilize complex information fusion to promote feature extraction, at the expense of efficiency. Currently, multiscale feature fusion has been widely used as an effective means to boost the feature extraction capability of encoding–decoding CNNs, and more sufficient information flow utilization and more complex feature fusion approaches prevail [28]–[30]. However, the feature maps in the encoding and decoding layers of the mainstream methods are too thick, whose number of channels can be hundreds or thousands at every turn. It leads the multiscale feature fusion to consume computing power and storage and impose a great burden on efficiency, thus being difficult to use for routine deployment in industrial scenarios.
3) Common decoding upsampling approaches face certain drawbacks when dealing with RS images, which are not conducive to the final accuracy. Generally, decoding upsampling is one of the main difficulties and bottlenecks limiting the accuracy of prediction. At present, the most widely used decoding upsampling approaches are interpolation (such as the nearest neighbor, bilinear, etc.) and transposed convolution. However, the spatial resolution of RS images is relatively low and they are generally orthophotos, so the semantic association and contextual information between pixels are relatively weak. Therefore, the effect of interpolation upsampling whose response coverage is only in the subpixel neighborhood is somewhat limited. Although transposed convolution can make up for this defect by adjusting the receptive field, it comes with a large number of parameters especially if a large kernel size is needed, and the same kernel acts on the entire image which ignores the response to local variations [31]. In addition, decoding upsampling is also an ill-posed problem, how to effectively recover the lost low-level details from the coarse-grained high-level features is challenging.

Addressing the above issues, this work aims to propose a robust DSM regression model that can be trained as easily as a classification task and pursues the common improvement of its accuracy and efficiency.

1) A semantic joint DSM regression framework is proposed for monocular RS images, whose working mechanism is revealed in Fig. 1. The framework performs SS classification and DSM regression simultaneously in a joint multitask manner, while the multitask learning enables the shared backbone to extract complementary features to achieve better performance than each independent objective. In particular, a two-stage training objective of joint SS classification loss and DSM regression loss (joint SD loss) is introduced. The joint SD loss makes the classification task dominate the training in the early stage to achieve rapid convergence of the model and allows the regression task to participate as the model converges and take over the training in the mid-term to accomplish effective DSM prediction, and finally makes the two synergistic in the later stage to enhance the robustness and generalization performance of the model.
2) A full-scale feature multiplexing (FFM) encoding–decoding backbone is designed to realize model lightweight. The FFM backbone focuses on the unique perspective of feature reuse which achieves information decoupling through combined invocation and repeated utilization of the encoded states, thereby greatly

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reducing the number of channels of feature maps in all encoding and decoding layers, and the model efficiency can be improved accordingly.

3) A feature inpainting upsampling (FIU) module is introduced to improve the decoding upsampling accuracy. The FIU module aggregates multiscale features with rich information and reorganizes them into large receptive-field and locally discriminative dynamic kernels, which are performed convolution operations with the interpolation upsampling features to inpaint lost fine-grained information. The dynamic kernels are generated in an on-the-fly manner so as not to increase the number of model parameters.

In summary, there are three main contributions of this article.

1) An easy-to-train and robust monocular RS image semantic joint DSM regression framework is proposed along with its two-stage training objective.
2) An FFM backbone is designed to lighten the model and improve efficiency.
3) An FIU module is introduced to achieve the improvement of decoding as well as regression accuracy.

The rest of this article is organized as follows. Section II introduces the related works. The proposed methodology is elaborated in Section III. Experimental results are illustrated in Section IV. Finally, Section V concludes this article.

II. Related Works

This section briefly introduces some SOTA methods involving joint learning of height and semantic information from monocular RS images. Related methods mainly present two types of applications.

First, the joint multitask of height estimation and SS makes the two benefit each other to achieve better performances compared with a single task. In particular, Srivastava *et al.* [16] proposed a multitask CNN to perform SS and DSM prediction at the same time. This is one of the earlier studies of this issue, so its encoding–decoding network is relatively traditional (still using fully connected layers as outputs). Carvalho *et al.* [17] presented a multitask CNN that simultaneously estimates both height and semantic maps from a single-view RS image, and the backbone is vanilla fully convolutional architecture with skip connections. Zheng *et al.* [23] proposed an encoder–dual decoders architecture for end-to-end multitask learning. The dual decoders are two cascaded top-to-down feature pyramids to do SS and height regression, respectively, and the semantic features are leveraged to help height estimation. Kunwar [24] presented a method of model ensemble and transfer learning to predict semantic labels and DSM from a single-view RS image. First, an ensemble of different models (varied backbones employed in a U-net architecture) with different bands (specific band combination from the multispectral image) is trained for SS. Then, a coarse map is generated based on the height statistics of specific categories in the training samples and used as an additional band. Finally, an ensemble of different models (reuse weights trained by SS) with and without the coarse-map band is trained for height estimation. The above methods are effective and have achieved SOTA performances. However, [16] and [17] adopt transposed convolution for upsampling, which leads to a large number of parameters in their models. Meanwhile, the same kernel acts on the global feature maps will restrict the model's capability of responding to local variations. Furthermore, the training objectives of [16], [17], and [23] make the complicated and unstable regression task participate from the beginning, so the robustness of their model training is more or less arguable. The ensemble method of [24] achieves the stability and rapid convergence of the regression training as well as outstanding test accuracy, but its complex method flow (which requires more than ten times of separate training) is hard to deploy in industrial scenarios.

Second, the height information is embedded or integrated to improve the accuracy of SS. For example, Audebert *et al.* [18] proposed using DSM as *a priori* or posterior knowledge to perform multiscale fusion with multispectral RS data for improving the SS accuracy. Through their investigation, late fusion makes it possible to recover errors steaming from ambiguous data, while early fusion allows for better joint-feature learning but at the cost of higher sensitivity to missing data. Liu *et al.* [21] proposed to implement SS by distilling height-aware embeddings implicitly, and a modality-affinitive propagation block was presented to realize the semantics and height information fusion. In addition, a context-aware reorganization module was introduced, which assigns discriminative global context features to local positions. Li *et al.* [22] added a separate decoding branch to the encoding–decoding SS network, which is used to distill the height geometric features under the supervision of ground-truth DSM images. The height geometric features are fused with the contextual features from the SS branch for enhancing the SS performance. The above methods have proposed unique ideas and achieved SOTA results. However, first, the DSM they generate is not concerned with accuracy but only as an intermediate product for assisting SS, which is difficult to be put into the application as a qualified data product. Second, they also suffer somewhat from the model heaviness and upsampling handicap mentioned in Section I.

III. Proposed Mothed

For CV tasks, since related tasks may share the visualization characteristics of information, having the same model learn shared representations between different tasks is not only more efficient in terms of data, storage, and inference speed, but also enhances the robustness and generalization performance of the model [32]. Based on this, a semantic joint DSM regression model is proposed in the form of multitask learning to address the issue raised in this article.

A. Overall Framework

The overall framework of the proposed model is depicted in Fig. 1. Specifically, monocular RS image is converted into embeddings containing high-level semantic information and height information through the encoding–decoding backbone. En^1 to En^5 and De^1 to De^4 denote the encoding and decoding layers of the backbone, respectively. "*s*/*m*, *n*" indicates the

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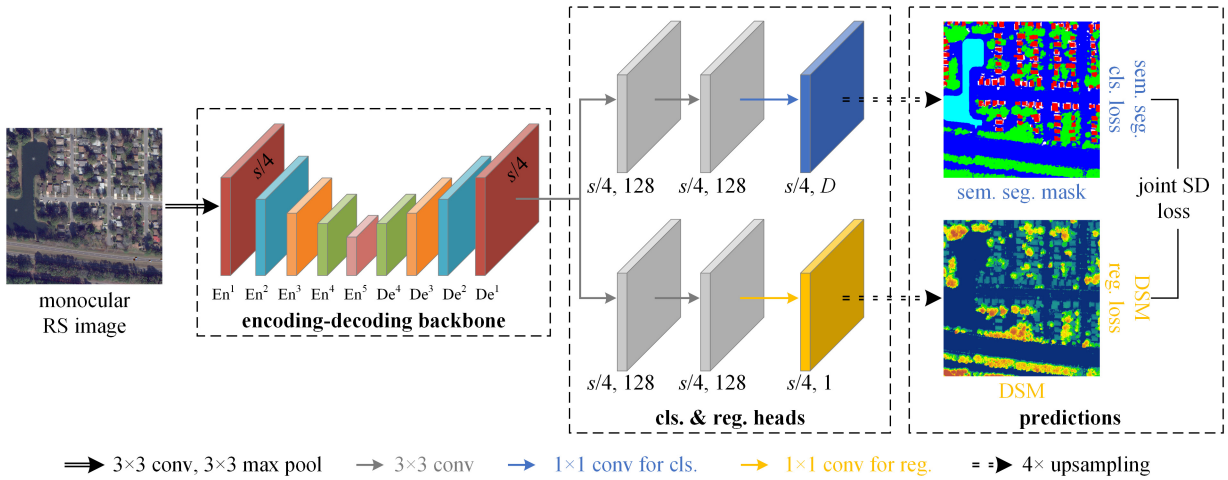


Fig. 1. Overall framework of semantic joint DSM regression model. En^1 to En^5 and De^1 to De^4 denote the encoding and decoding layers of the backbone, respectively. "s/m, n" indicates the size and number of channels of the feature map, where s refers to the scale of the input size, m refers to the downsampling ratio, and n refers to the number of channels. D denotes the number of semantic categories.

size and number of channels of the feature map, where s refers to the scale of the input size, m refers to the downsampling ratio, and n refers to the number of channels. SS classification and DSM regression are performed simultaneously in the dual-branch output heads. Each location of the classification final layer predicts a D-dimensional vector, where D denotes the number of semantic categories. Each location of the regression final layer predicts a height value. Note that the size of the final layers is one-fourth of the size of the input monocular RS image, so the final layers undergo $4\times$ interpolation upsampling to generate the SS and DSM predictions. The joint SD loss will be elaborated in Section III-D, and additional details are illustrated in Fig. 1.

B. FFM Encoding–Decoding Backbone

Many excellent studies have confirmed that multiscale feature fusion is a powerful tool to boost the capability of encoding–decoding SS networks, but their thick feature maps lead to huge computational complexity, as stated in Section I. For the purpose of thinning the thickness of feature maps to achieve lightweight multiscale feature fusion, this work proposes an elegant full-scale feature-multiplexing encoding–decoding backbone, which is inspired by the core idea of DenseNet [33] that drastically reduces the number of feature map channels through feature reuse. The architecture of the FFM encoding–decoding backbone is illustrated in Fig. 2. Specifically, each encoding layer and decoding layer consists of three and two feature maps, respectively, which is a reference to the classic U-net [28]. Fig. 3 demonstrates En^3 and De^3 as examples, and other encoding and decoding layers are in the same form. The last feature maps in the encoding (decoding) layers are denoted as en^1 to en^5 (de^1 to de^4), and the first feature maps in the decoding layers are denoted as de^{1f} to de^{4f}. "s/m, n" in Fig. 2 refers to the size and number of channels of the en^x or de^x. en^1 to en^4 are repeatedly aggregated into full-scale features and leveraged to various decoding stages. en^5 is not aggregated into the fusion features since it also serves as the initial state of the decoding stage. The downsampling operation in Fig. 2 is 3×3 conv, and the upsampling operation adopts bilinear interpolation. FIU-1 to FIU-4 are the iterative upsampling modules for generating de^{1f} to de^{4f}, respectively.

The FFM backbone realizes information decoupling and model lightweight through feature reuse, which is a relatively unique perspective rarely focused on by mainstream SS networks. Taking the classic U-net as an example, U-net is generally a one-way information flow (except for a few skip connections). Each of its encoding and decoding layers contains previously complete information and can be used as an independent global state of the network, which is similar to the Markov process. That is, the next state of the network is derived only from the current global state and has nothing to do with the previous states. This fashion causes the previous encoding and decoding information to be replicated in later layers, and this article believes it is the reason they have to be so thick. In contrast, the FFM backbone invokes all previous encoded states instead of relying on a single global state for decoding, so the encoding information can be decoupled from the decoding layers, and the encoding information of previous layers can be decoupled from later layers, thereby being able to make the encoding and decoding layers thin. Compared to U-net, the number of channels of the corresponding feature maps in the FFM backbone is reduced by about 75% (e.g., the en^1 in U-net has 64 channels, and the en^5 in U-net has 1024 channels), thus being quite lightweight. As for how the number of channels in each layer of the FFM backbone is determined, it is indeed an empirical attempt with qualitative analysis. It is well known that the network structure hyperparameters of CNNs are hard to obtain by quantitative experiments unless resorting to reinforcement learning (such as neural architecture search) and relying on extremely large computing power, which is currently a hardware limitation for this work.

C. Feature Inpainting Upsampling Module

The FIU module works as an upsampling and channel compression operator with variable receptive field and locally discriminative dynamic kernels, and Fig. 4 takes FIU-2 as

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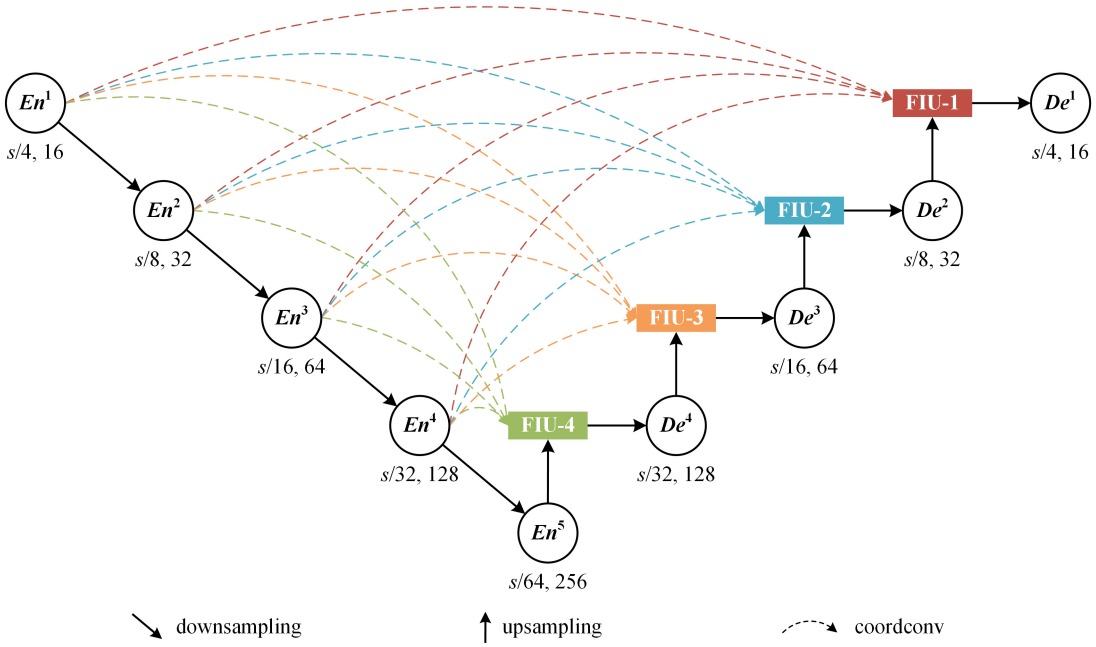


Fig. 2. Architecture of the FFM encoding–decoding backbone. FIU-1 to FIU-4 are the iterative upsampling modules for generating de^{1f} to de^{4f}. Downsampling is 3 × 3 conv, and upsampling is bilinear interpolation. Coordconv is the same as in [34]. "*s*/*m*, *n*" refers to the size and number of channels of the en^x or de^x.

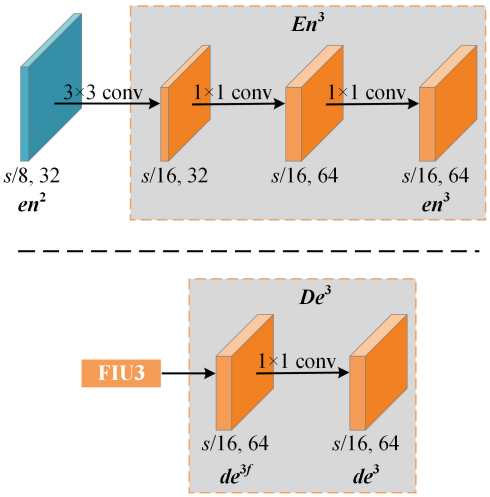


Fig. 3. Internal structures of En^3 (upper) and De^3 (lower).

an example to illustrate its working mechanism. Specifically, the input to the FIU module consists of two parts. The first part is the interpolation input, which is de^3 upsampled by a factor of 2 using bilinear interpolation. The second part is the coordconv input. en^1 to en^4 are resized to the same size as the interpolation input without changing their number of channels by coordconv operation [34]. In particular, two additional channels which contain normalized locations x and y coordinates are concatenated to the original feature map to encode spatial information; 3 × 3 conv and transposed conv are adopted when the scaling factor is greater and less than 1, respectively; appropriate stride and padding are chosen to obtain the expected feature map size. The coordconv input feature maps are first concatenated and then performed two 1 × 1 convs to be reorganized into the full-scale spatial (FSS) feature map. Note that two feature maps of the same shape are actually generated after two convs, but only the latter one is shown in Fig. 4. The FSS feature map is the combination of the location-wise dynamic convolution kernels. For a location $(i,\ j)$ in the FSS feature map which is a $2\sigma^2$-dimensional vector in the channel direction (supposing σ is an integer, and FIU-2 is illustrated with $\sigma = 3$ in Fig. 4), it can be seen as consisting of two $\sigma \times \sigma$ kernels, where the channels $[1, \sigma]$ and $[\sigma + 1, 2\sigma]$ are transposed to the first and second ones, respectively. For the same location $(i,\ j)$ in the interpolation input, it is divided into two by two groups. Two $\sigma \times \sigma$ single-channel subregions centered on the two elements in each group are cropped from the interpolation input. The two subregions are sequentially performed convolution operations with the two kernels at the same location in the FSS feature map, and the results are added and then given as the corresponding element in location $(i,\ j)$ of the de^{2f}.

The FIU module aggregates full-scale encoded features and reorganizes them into the FSS features, and the feature vector along the channel direction at each location contains the rich full-scale information of this local region. Each location feature vector is transposed into two dynamic convolutional kernels of variable size, so it can also act as a 1/2 channel compressor while saving a layer of channel convolution. At the same time, the upsampling receptive field can be increased accordingly by appropriately adjusting the kernel size σ. After dedicated testing, this article finds that better results can be achieved when $\sigma = 5$, which is also adopted in the subsequent experiments. The feature inpainting of the FIU module is reflected in two aspects, where the full-scale encoded information is employed to compensate for 1) the disregarded contextual content due to interpolation upsampling being limited to subpixel neighborhood and 2) the fine-grained

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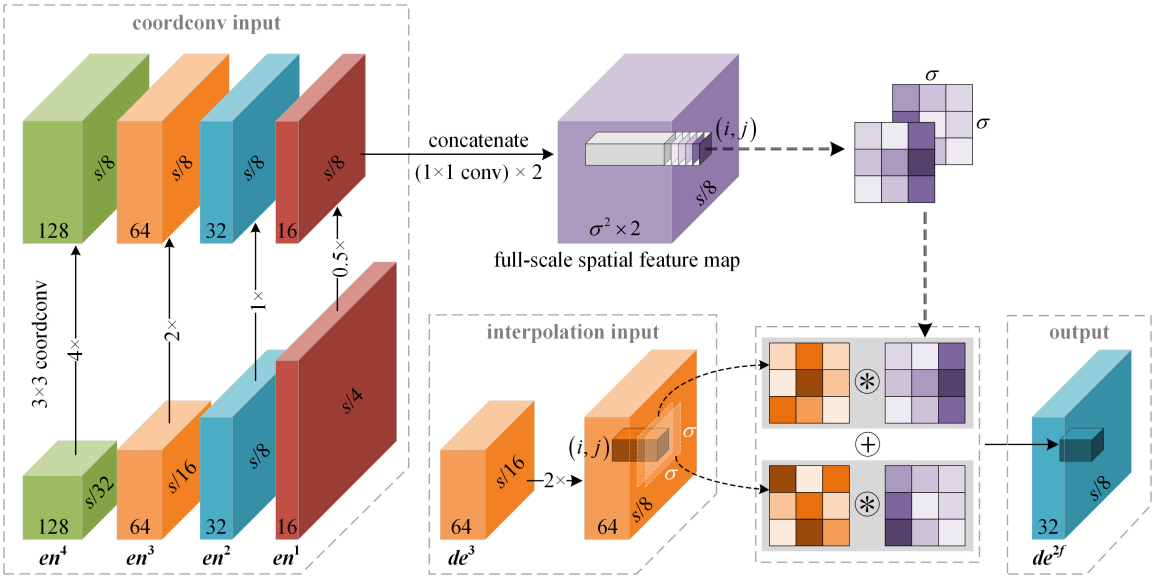


Fig. 4. Structure of the FIU-2, which is used to upsample de^3 to de^{2f}. $\circledast$ and $\oplus$ represent convolution and addition operations, respectively. The FIU-2 is illustrated with $\sigma = 3$.

information lost in the high-level coarse-grained feature maps. Additionally, the dynamic kernels are predicted on-the-fly rather than learned as network parameters, so compared to transposed convolution, it is very beneficial to model lightweight. Furthermore, predicting specialized dynamic kernels for different locations instead of using the same kernel to act on the whole allows local variations to render better.

D. Training Objective

The proposed two-stage training objective is defined as follows:

$$L_{\mathrm{SD}}(N) = \begin{cases} L_{\mathrm{cls}}(N), & \text{if } \psi(T) < \alpha, \quad \text{stage I} \\ e^{L_{\mathrm{cls}}(N)} + L_{\mathrm{reg}}(N), & \text{else}, \quad \text{stage II} \end{cases} \tag{1}$$

where N and T indicate the current iteration index and epoch index, respectively; L_{SD}, L_{cls}, and L_{reg} refer to the joint SD loss, SS classification loss, and DSM regression loss, respectively; conventional focal loss [35] is adopted as L_{cls} and smooth L1 loss is adopted as L_{reg}; ψ is the switching function of the two training stages, which is defined as follows:

$$\psi(T) = \frac{k \cdot \min_{(T-k) \le t \le T-1} L_{\mathrm{acls}}(t)}{\sum_{t=T-k}^{T-1} L_{\mathrm{acls}}(t)} \tag{2}$$

where L_{acls} is calculated as the average L_{cls} in each epoch over several iterations, and the degree of change of L_{acls} in the last k epochs determines the value of $\psi(T)$. Simply put, the meaning of $\psi(T)$ is that if the L_{acls}s of the previous consecutive k epochs have changed little, the model is considered to have converged on the classification task and can switch to stage II to start training the regression task. α in (1) is the threshold to measure the degree of change of the L_{acls}s, and according to the follow-up experiments, this article chooses $k = 5$ and $\alpha = 0.975$. When $T \le k$, let $\psi(T) = 0$. Note here that the switch between the two stages of the joint SD loss is one-shot and irreversible. In fact, the first iteration of each epoch will make a judgment, once the joint SD loss enters stage II, it will be maintained until the training is terminated.

The strategy of joint SD loss reflects the relationship between the SS classification and height regression branches, while the two are similar to pretraining and fine-tuning for transfer learning, respectively, but pretraining runs throughout. Specifically, the semantic category of a pixel is related to the physical meaning of its height value, and the visualization characteristics of the semantic label and the DSM are consistent. Therefore, it is logically reasonable to use the relatively easy-to-train SS classification model as the initialization parameters for the difficult height regression model training, and thus stage I solely adopts L_{cls} as the training objective while making the model converge quickly. Stage II starts when the model state triggers the switch condition, and the height regression task begins to participate in training. The SS task is still functional at stage II, in order to make the complicated continuous predictions subject to discretization constraints, so that the prediction errors fluctuate within reasonable intervals, thereby greatly reducing the appearance of outliers and effectively strengthening the training robustness. L_{cls} is taken as a base-e exponent so that it is on the same order of magnitude as L_{reg} without being overwhelmed. In the early stage of stage II, L_{reg} is relatively large and $e^{L_{\mathrm{cls}}}$ is close to a constant, and the training is dominated by the regression task; in the later stage, L_{reg} has also converged and close to $e^{L_{\mathrm{cls}}}$, so the two play almost equal roles until the training is completed, which is what the training ultimately expects.

IV. Experiments

In this section, the proposed method is tested and compared with related works, experimental results are analyzed, and ablation studies are conducted.

A. Dadasets and Metrics

This article employs two well-known datasets to carry out the experiments, which are provided by the International Society for Photogrammetry and Remote Sensing (ISPRS) for

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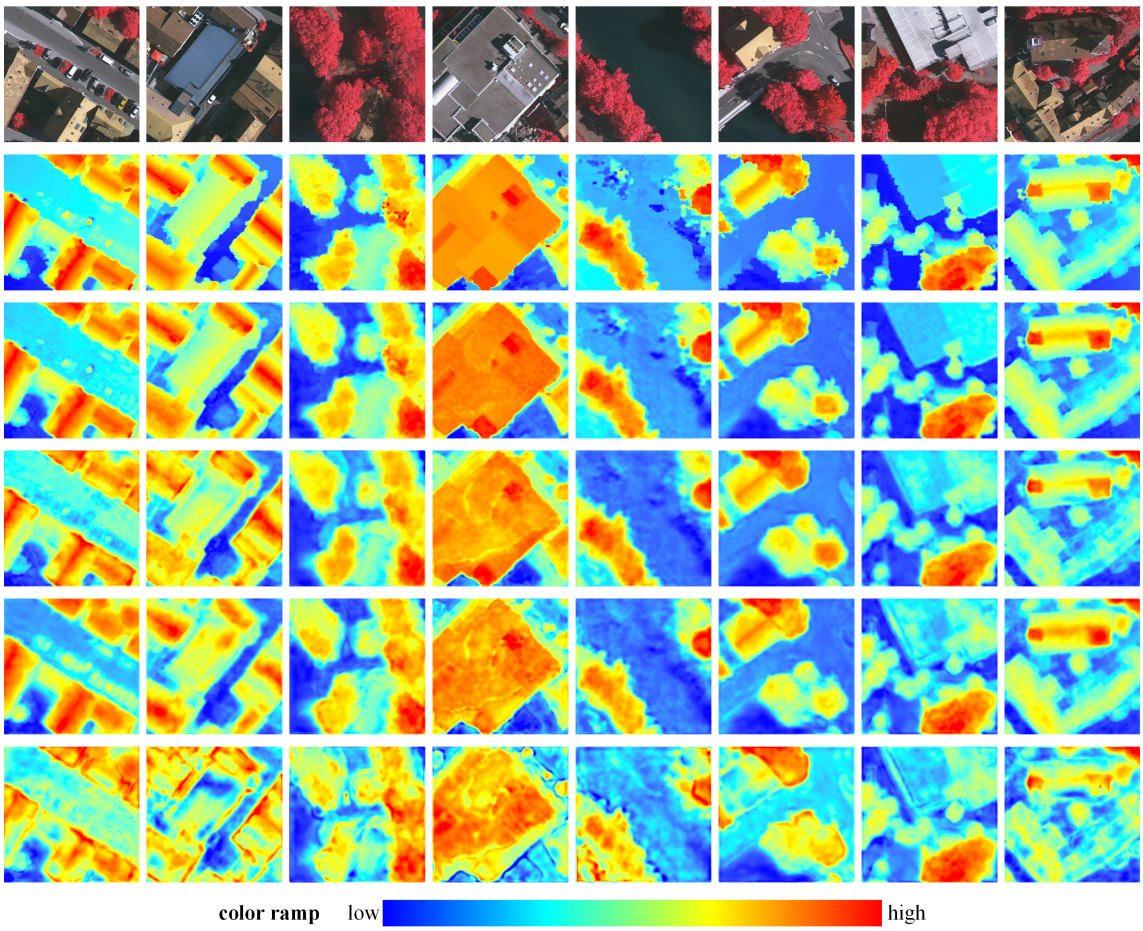


Fig. 5. Visualized DSM reconstruction test results of some small-scale local regions in Vaihingen. First row: RS images, second row: GTs, rows 3 to 6: DSM reconstructions of the baseline, DANet†, PSPNet†, and U-net†, respectively.

urban classification and 3-D building reconstruction: Vaihingen[1] and Potsdam.[2] Both dataset regions cover urban scenes where Vaihingen is a relatively small village and Potsdam is a typical historical city. Both datasets contain DSMs generated from high-resolution orthographic photographs and are manually annotated with semantic labels for the six common land cover categories: impervious surface, building, low vegetation, tree, car, and clutter/background, where clutter/background includes water and objects not concerned in urban scenes. In particular, Vaihingen contains 33 RS images of various sizes (about from 1900 × 1900 to 2500 × 3800 pixels), and the spatial resolution [ground sampling distance (GSD)] of the RS image and DSM is 9 cm. The RS image consists of three bands: near-infrared (NIR), red (R), and green (G). Potsdam contains 28 RS images of the same size, and each has 6000 × 6000 pixels and four bands: R, G, blue (B), and NIR. The DSM of Potsdam is produced using some automated filtering workflow without manual quality control, so there is no guarantee that there are no errors in the data. The GSD of the RS image and DSM in Potsdam is up to 5 cm. For each dataset, the RS images are cropped into 512 × 512 patches by a sliding window with a step size of 384, and the DSMs and semantic labels are processed accordingly to correspond to the patches. For Potsdam, RGB images are used and some data with errors in DSM are manually removed. Finally, 1566 and 4874 sets of samples on Vaihingen and Potsdam are obtained, respectively. 1/2, 1/6, and 1/3 of the data are randomly divided into training, validation, and test sets, respectively. Two numerical metrics, the root-mean-square error (RMSE) and the zero-mean normalized cross correlation (ZNCC), are used to evaluate the quality of the predicted DSM, which is the same as in [25]. In addition, the performance on SS of the proposed joint multitask model is also evaluated based on the F1 score and overall accuracy (OA), which are the same as in [21].

[1]https://www2.isprs.org/commissions/comm2/wg4/benchmark/2d-sem-label-vaihingen/, accessed on December 1, 2021.

[2]https://www2.isprs.org/commissions/comm2/wg4/benchmark/2d-sem-label-potsdam/, accessed on December 1, 2021.

B. Implementation Details

The baseline model in this article (semantic joint framework + FFM backbone + FIU module) is implemented based on MMSegmentation [36] under Python 3.6.10 and Pytorch 1.6.0. The provided U-net model trained on Cityscapes is adopted as the initial weight, and the newly added layers are initialized as in [35]. Stochastic gradient descent (SGD) is used as an optimizer, and its weight decay and momentum are set as 0.0001 and 0.9, respectively. The baseline model is trained for 180 epochs on both datasets with a batch size of 4. The initial learning rate is 0.01 and is reduced tenfold after 3/4 and 11/12 of the total epochs. Moreover, some SOTA

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TABLE I
QUANTITATIVE DSM RECONSTRUCTION TEST RESULTS ON VAIHINGEN

Methods	RMSE	ZNCC	Param. (M)	FPS	Time (h)
CycleGAN [37]	5.86±0.57	0.485±0.110	28.3	17.25*(R)*	24.85
U-IMG2DSM [19]	3.46±0.78	0.808±0.073	38.32	11.94*(R)*	16.62
Amirkolaee [26]	2.871	0.814±0.095*(R)*	34.70*(R)*	14.83*(R)*	5.33*(R)*
IMG2DSM [25]	2.58±**0.09**	0.813±**0.072**	57.18	9.21*(R)*	2.65
MPFusion [27]	2.42	0.858	46.54*(R)*	12.84*(R)*	6.03*(R)*
StyHighNet [20]	2.4	0.872	24.11*(R)*	21.78*(R)*	3.77*(R)*
DORNet [15]	1.698	0.826±0.079*(R)*	53.52*(R)*	9.96*(R)*	6.82*(R)*
U-net [28]†	3.064±0.552	0.811±0.127	31.05	16.56	11.71*(M)*
PSPNet [29]†	2.271±0.264	0.853±0.106	67.95	7.26	25.49*(M)*
DANet [30]†	1.907±0.115	0.845±0.081	68.84	6.97	26.51*(M)*
baseline	**1.645**±0.134	**0.877**±0.091	**8.97**	**49.31**	**1.13**
baseline‡	2.560±0.797	0.816±0.113	**8.97**	49.29	3.05

[1] † indicates that the model is transformed into the proposed joint multi-task framework.
[2] ‡ indicates that the classification and regression tasks are optimized in parallel.
[3] *(R)* indicates that the result is reproduced by this paper.
[4] *(M)* indicates that the model actually adopts multi-GPU distributed training, and the training time is converted to the single GPU case.

TABLE II
QUANTITATIVE DSM RECONSTRUCTION TEST RESULTS ON POTSDAM

Methods	RMSE	ZNCC	Param. (M)	FPS	Time (h)
CycleGAN [37]	6.79±1.05	0.477±0.180	28.3	17.29*(R)*	24.85
U-IMG2DSM [19]	4.62±1.24	0.740±0.124	38.32	11.90*(R)*	16.62
Amirkolaee [26]	3.468	0.788±0.147*(R)*	34.70*(R)*	14.74*(R)*	16.26*(R)*
IMG2DSM [25]	3.89±**0.11**	0.718±**0.008**	57.18	9.22*(R)*	**2.65**
MPFusion [27]	3.87	0.829	46.54*(R)*	13.02*(R)*	17.61*(R)(M)*
StyHighNet [20]	3.83	0.845	24.11*(R)*	21.82*(R)*	11.72*(R)(M)*
DORNet [15]	**1.404**	0.813±0.150*(R)*	53.52*(R)*	9.91*(R)*	19.64*(R)(M)*
U-net [28]†	3.505±0.702	0.797±0.134	31.05	16.52	36.45*(M)*
PSPNet [29]†	2.996±0.563	0.820±0.165	67.95	7.30	81.39*(M)*
DANet [30]†	2.383±0.221	0.836±0.089	68.84	6.99	87.02*(M)*
baseline	1.719±0.162	**0.856**±0.082	**8.97**	**49.27**	3.83

monocular RS image DSM reconstruction-related works are reproduced for comparison. Specifically, CycleGAN [37],[3] U-IMG2DSM [19],[4] IMG2DSM [25],[5] and StyHighNet [20][6] are reproduced through their open-source codes. Amirkolaee [26], MPFusion [27], and DORNet [15] are reproduced with reference to their original articles and based on MMSegmentation, where MPFusion's implementation also refers to project RefineNet.[7] For a fair comparison, and considering their original train-test splits are also random, the comparison methods are trained on the same datasets as the baseline. MPFusion is trained for 60 epochs with a batch size of 4, which is not specified in its original article, and the implementation details of the others are the same as in their original articles. Note here that the models of CycleGAN, U-IMG2DSM, and IMG2DSM are not trained in this article. This is because the purpose of the comparison method reproductions is to complete the concerned experimental metrics of this article which are not tested in their original articles. Paoletti *et al.* [19] has given all the concerned experimental metrics of CycleGAN, U-IMG2DSM, and IMG2DSM except the frames per second (FPS), and only testing the FPS does not need to actually train the models. The experiments in this article are running on ServMax TS40-X2 Tower Workstation with Intel Xeon Gold 6226R CPU, NVIDIA GeForce RTX 3090 24-GB GPU, and 256-GB memory.

[3]https://github.com/junyanz/pytorch-CycleGAN-and-pix2pix, accessed on May 11, 2022.
[4]https://github.com/mhaut/UIMG2DSM, accessed on May 11, 2022.
[5]https://github.com/joseph-zhang/img2dsm-pytorch, accessed on May 11, 2022.
[6]https://github.com/gagiter/StyHighNet, accessed on May 11, 2022.
[7]https://github.com/DrSleep/refinenet-pytorch, accessed on May 11, 2022.

C. Results

1) DSM Reconstruction: The quantitative test results of DSM reconstruction on Vaihingen and Potsdam by the baseline as well as the comparison methods are reported in Tables I and II, respectively. For the experimental data of the comparison methods, *(R)* indicates that it is the result reproduced by this article, while the others are from their original articles. Since the reproduced results are usually not as good as the original results, for the objectivity of the evaluation, this article only presents the reproduced results for the missing indicators. Moreover, the original experimental data of CycleGAN, U-IMG2DSM, and IMG2DSM are all from the [19]. Bold items in the tables represent the best results on the corresponding metrics. In terms of accuracy, the values before and after "±" in RMSE and ZNCC represent the mean and maximum variance of the test results, respectively. Taking the mean as the main consideration, the results confirm that among the four accuracy metrics on the two datasets, the baseline method achieves three optimal and one suboptimal. In particular, its RMSE on Vaihingen is reduced by about 3.1% compared with the SOTA method, and its ZNCCs are improved by 0.5% and 1.1% compared with

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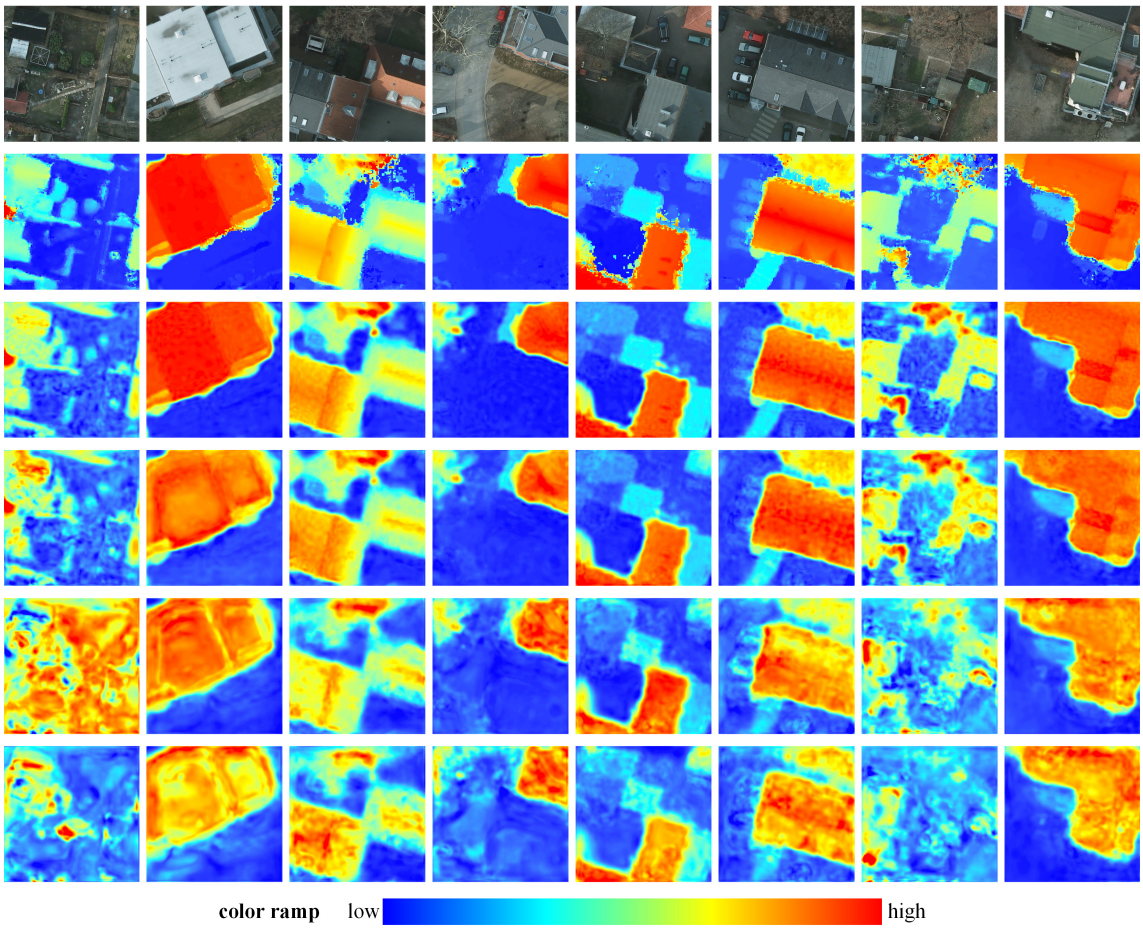


Fig. 6. Visualized DSM reconstruction test results of some small-scale local regions in Potsdam. First row: RS images, second row: GTs, rows 3 to 6: DSM reconstructions of the baseline, DANet†, PSPNet†, and U-net†, respectively.

the SOTA methods on Vaihingen and Potsdam, respectively. In terms of efficiency, the model parameters (millions), FPS at inference, and total training time (hours) of the methods are counted separately. It is salient that the baseline model is quite lightweight, and even compared to the second-lightest model, it still has nearly 63% fewer parameters. Accordingly, its inference FPS and training time are also far superior to other methods. It seems that the training time of IMG2DSM on Potsdam is shorter than the baseline. However, the training data of IMG2DSM are 256 × 256 pixel size, while those of the baseline are 512 × 512; in addition, the training set of IMG2DSM has only 400 images, while the baseline has more than six times as many training samples. Therefore, the training efficiency of the baseline on Potsdam should also be optimal. The visualized DSM reconstruction test results of the baseline are depicted in Figs. 5–8. In particular, Figs. 5 and 6 cover some individual scenes on Vaihingen and Potsdam, respectively, including buildings, trees, streets, low vegetation, water, and so on, where the first three rows are the RS images, DSM ground truths (GTs), and DSM reconstructions of the baseline, respectively, and cool to warm colors in DSM correspond to low to high height values. Figs. 7 and 8 cover a large area of Vaihingen and Potsdam, respectively, for the overall presentation. In particular, Fig. 7 is a part of three adjacent areas $4^{\#}$, $5^{\#}$, and $26^{\#}$ of Vaihingen, and Fig. 8 is a part of three adjacent areas $10^{\#}$, $11^{\#}$, and $12^{\#}$ of Potsdam. Fig. 7 has a sense of patchwork because the images in Vaihingen are not seamlessly stitched. The overall reconstructed DSMs are overlaid on the RS images in translucent mask fashions, and the correspondence between ground objects and height values is more clearly displayed. According to the visualized results, whether in terms of overall or detail, in terms of numerical accuracy or ground object distribution structure similarity, the DSMs reconstructed by the baseline intuitively exhibit outstanding and reliable quality. In summary, this method is effective and efficient in DSM reconstruction for monocular RS images. Not only does it achieve better or comparable accuracy to SOTA-related methods, but it is also very lightweight and efficient so as easy to deploy and apply in engineering scenarios.

2) Semantic Segmentation: Experiments in this section also investigate the performance of this method on SS of RS images. Some SOTA-related works which also take DSM as input for SS are cited for comparison, and the quantitative test results on Vaihingen and Potsdam are shown in Tables III and IV, respectively, where the results of the comparison methods are from [21]. The F1 score for each semantic category, mean F1 score, and OA are counted separately. It can be seen from various indicators that the baseline is able to realize high-quality SS, while achieving effective DSM reconstruction. Even compared to SOTA methods dedicated to RS SS, its accuracy is still comparable. Figs. 9 and 10

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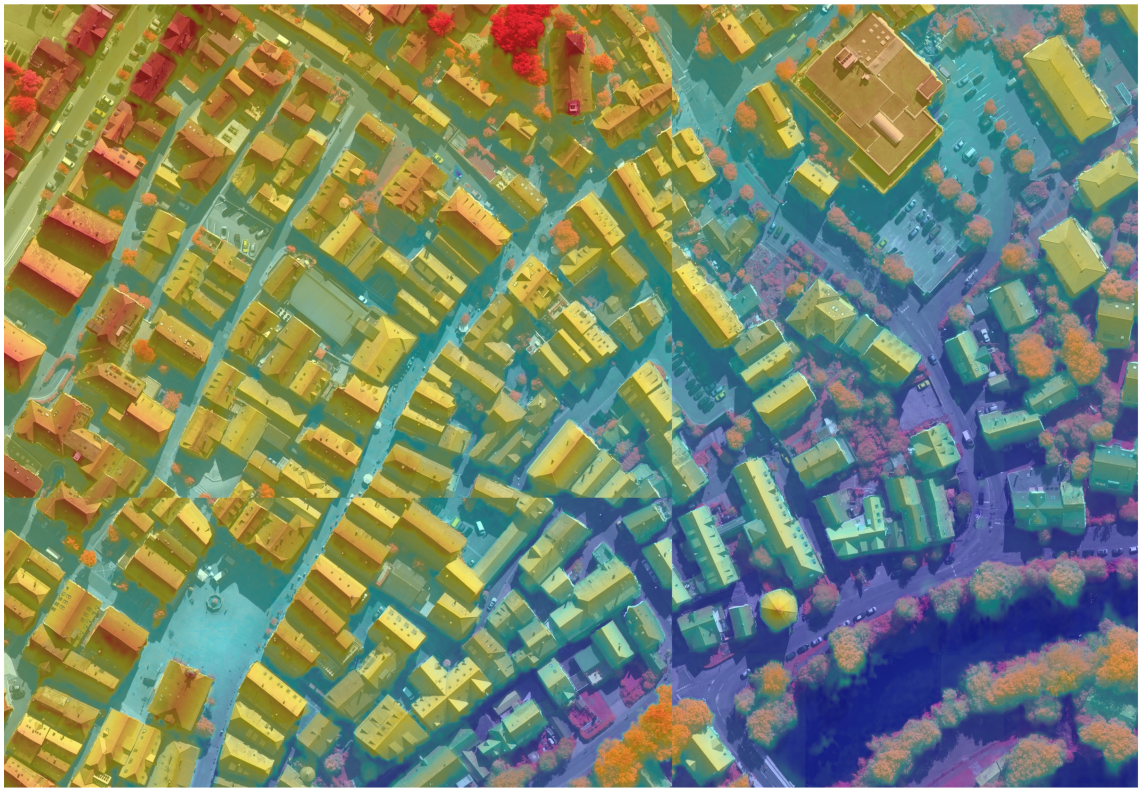


Fig. 7. Visualized DSM reconstruction test result of the baseline on a large-scale overall region in Vaihingen.

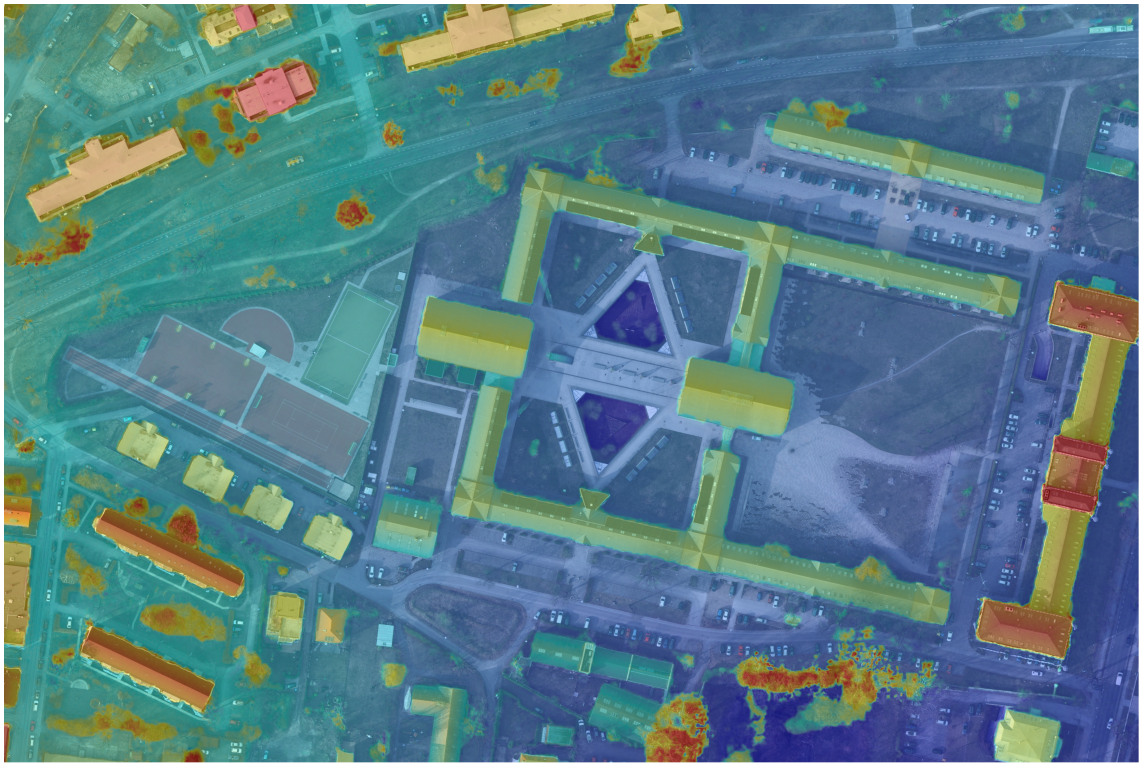


Fig. 8. Visualized DSM reconstruction test result of the baseline on a large-scale overall region in Potsdam.

are the visualized SS test results on Vaihingen and Potsdam, respectively. The first three rows in the figures are RS images, GTs, and test results of the baseline, respectively. Both figures clearly demonstrate the accuracy and validity of the proposed method in this regard.

D. Ablation Studies

1) Role of the Joint Multitask Framework: The effect of the proposed framework on the convergence and robustness of model training is investigated in this section, and three SOTA SS models U-net [28], PSPNet [29], and DANet [30]

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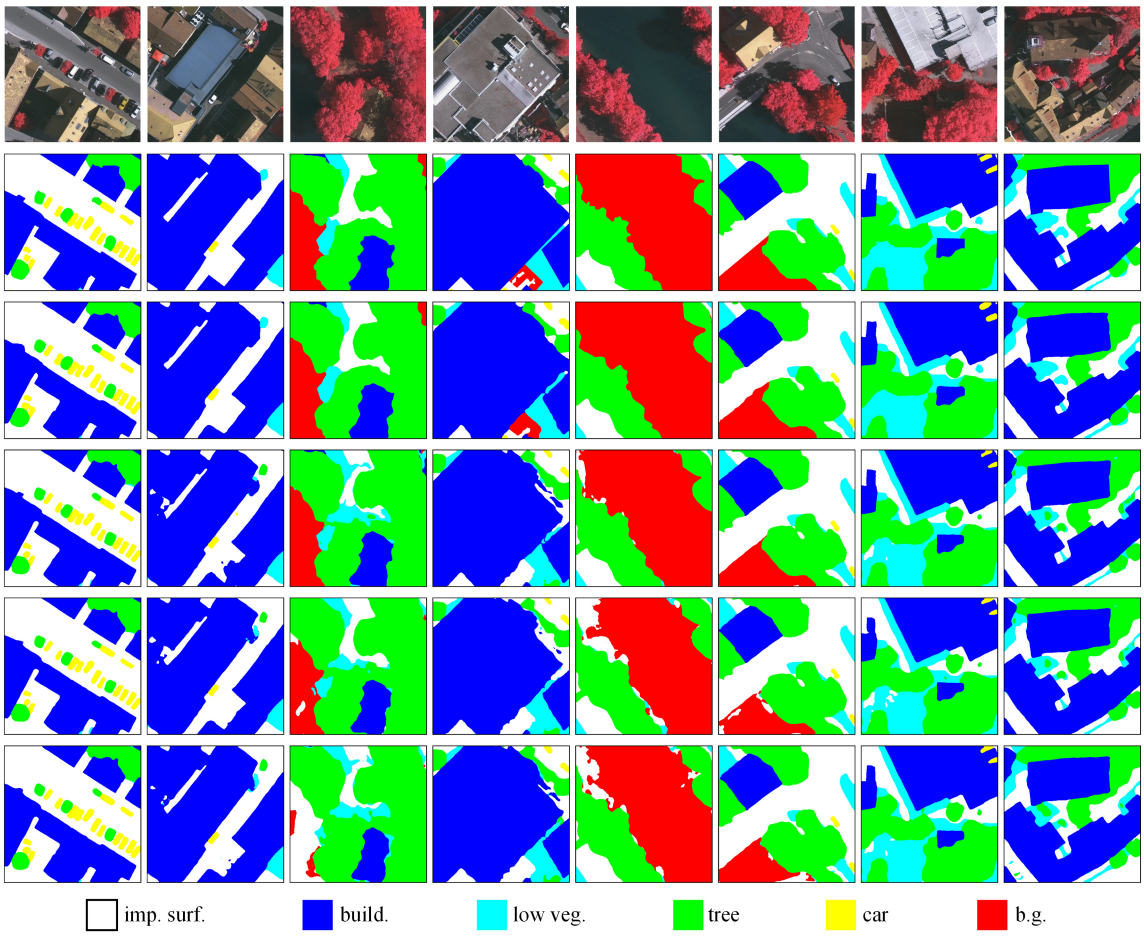


Fig. 9. Visualized SS test results on Vaihingen. First row: RS images, second row: GTs, rows 3 to 6: SS predictions of the baseline, DANet†, PSPNet†, and U-net†, respectively.

TABLE III
QUANTITATIVE SS TEST RESULTS ON VAIHINGEN

Method	F1					Mean F1	OA
	Imp. Surf.	Build.	Low veg.	Tree	Car		
UZ [2]	89.2	92.5	81.6	86.9	57.3	81.5	87.3
RoteEqNet [38]	89.5	94.8	77.5	86.5	72.6	84.18	87.5
ONE [39]	91	94.5	84.4	89.9	77.8	87.5	89.8
V-FuseNet [18]	92	94.4	84.5	89.9	86.3	89.42	90
DLR [40]	92.4	95.2	83.9	89.9	81.2	88.52	90.3
TreeUNet [41]	92.5	94.9	83.6	89.6	85.9	89.3	90.4
BKHN [21]	92.9	96	84.6	89.9	88.6	90.4	91
HECR-Net [21]	**93.64**	95.53	**85.78**	**90.37**	**89.09**	**90.89**	**91.45**
U-net [28]	87.89	90.59	79.42	81.5	64.57	80.79	85.89
PSPNet [29]	90.97	94.3	83.31	87.94	79.01	87.11	88.85
DANet [30]	91.36	93.62	83.35	88.78	82.71	87.96	89.14
U-net [28]†	88.11	91.23	78.75	83.38	66.7	81.63	86.55
PSPNet [29]†	91.19	93.86	84.25	88.62	81.41	87.87	89.17
DANet [30]†	90.92	94.76	84.87	88.51	84.95	88.8	89.6
baseline	92.86	**96.21**	84.83	89.77	88.65	90.46	91.18
baseline‡	91.63	94.89	84.90	87.42	83.38	88.44	89.23

are employed for validation and comparison. Specifically, first, they are directly modified into regression manners. For each of them, the last classification activation layer (softmax) is removed, and two layers of channel convolution are added after the SS output layer to output a single-channel regression layer. For comparison, the U-net, PSPNet, and DANet are also transformed into the proposed joint multitask frameworks. For each of them, another branch is drawn from the feature map before its SS output layer, and two layers of channel convolution are appropriately added to get a new output layer for simultaneous DSM regression, which is similar to the head in Fig. 1. The models modified directly into the regression manner and transformed into the joint multitask framework are denoted as "*" and "†," respectively. U-net*, PSPNet*, and DANet* are trained on Vaihingen for 720 epochs with a batch size of 16, an initial learning rate of 0.0001, and the smooth L1 loss. They are not trained on Potsdam because it is later confirmed that they cannot converge on Vaihingen,

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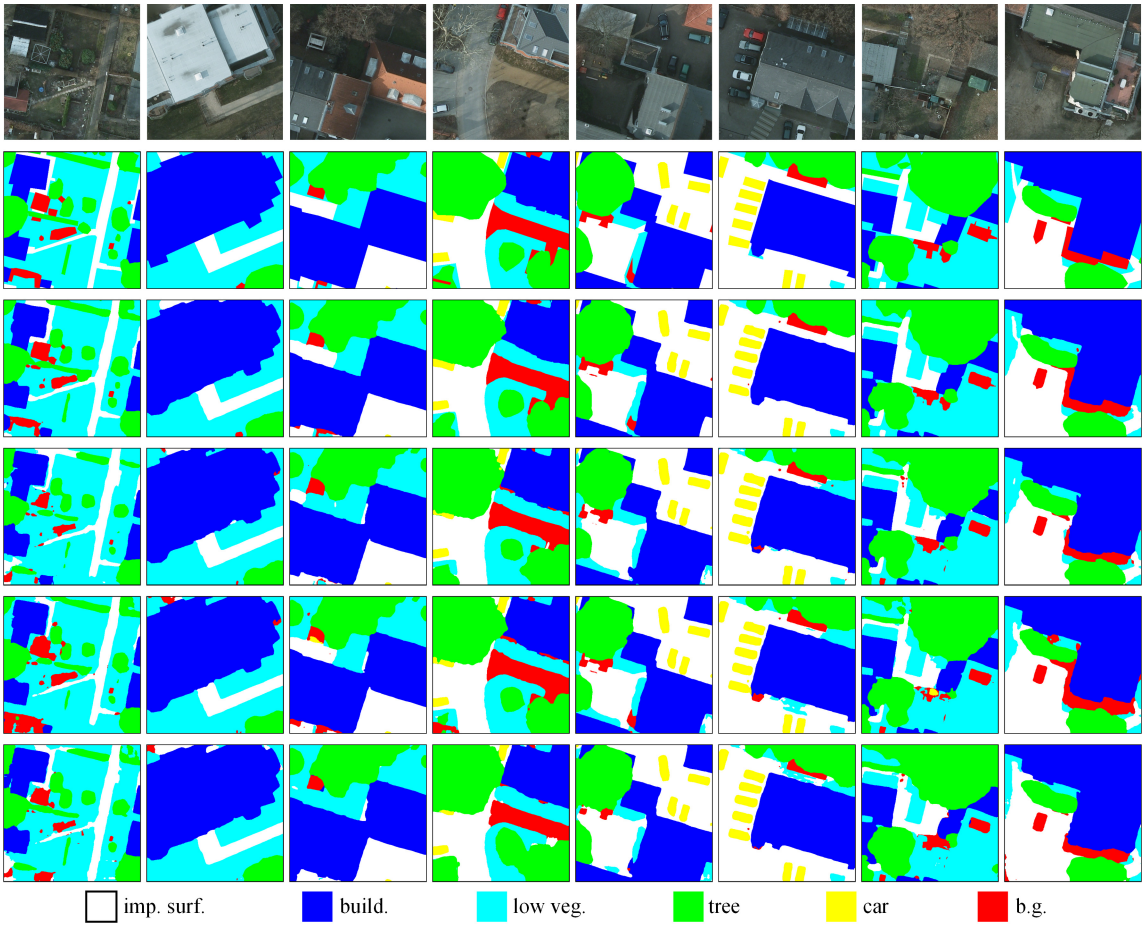


Fig. 10. Visualized SS test results on Potsdam. First row: RS images, second row: GTs, rows 3 to 6: SS predictions of the baseline, DANet†, PSPNet†, and U-net†, respectively.

TABLE IV
QUANTITATIVE SS TEST RESULTS ON POTSDAM

Method	F1					Mean F1	OA
	Imp. Surf.	Build.	Low veg.	Tree	Car		
UZ [2]	89.3	95.4	81.8	80.5	86.5	86.7	85.8
V-FuseNet [18]	92.7	96.3	87.3	88.5	95.4	92.04	90.6
TreeUNet [41]	93.1	97.3	86.6	87.1	95.8	91.98	90.7
BKHN [21]	93.3	97.2	88	88.5	**96**	92.6	91.1
HUSTW [42]	93.6	**97.6**	88.5	88.8	94.6	92.62	91.6
HECR-Net [21]	**93.75**	97.44	**88.68**	**89.19**	95.4	**92.89**	**91.84**
U-net [28]	88.65	93.72	79.79	81.42	73.16	83.35	86.21
PSPNet [29]	90.73	93.63	82.94	85.76	81.35	86.88	89.1
DANet [30]	91.27	95	84.36	85.92	84.13	88.14	90.23
U-net [28]†	88.56	94.09	80.51	81.77	77.35	84.46	86.54
PSPNet [29]†	91.22	94.28	82.69	86.3	83.4	87.58	89.75
DANet [30]†	91.18	95.91	86.05	86.27	84.45	88.77	90.36
baseline	93.48	97.28	88.43	88.85	95.57	92.72	91.67

so there is no need to train them on a larger dataset. U-net†, PSPNet†, and DANet† are trained on Vaihingen and Potsdam for 180 epochs with a batch size of 4, the initial learning rate of 0.01, and the joint SD loss. All the above models are implemented based on MMSegmentation, suitable initial weights are chosen from the model zoo, and other details are the same as the baseline method. The loss value curves of model*s and model†s (including the baseline) trained on Vaihingen are plotted in Fig. 11(a) and (b), respectively. The x-axes in the figures represent the iterations, and for Vaihingen, 40 000 iterations is about 180 epochs when the batch size is 4. For visual effects, 400 points are sampled at intervals of 100, and the loss values greater than 16 in Fig. 11(a) and 10 in Fig. 11(b) are truncated, respectively. It is obvious from Fig. 11(a) that the loss values of model*s oscillate violently and hardly converge, even though a fairly small learning rate, a large batch size, and a large number of epochs have been set (under the same iterations, since the batch size for model* training is four times that of model†, so the epochs are correspondingly four times that of model†). In fact, the robustness of model* training is rather poor, and if the above tricks are not adopted, gradient explosion will easily occur and the

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training will not be sustainable. In contrast, Fig. 11(b) shows that the loss values of model†s are generally able to converge effectively despite small oscillations. The step jump of each curve in the figure indicates that the joint SD loss switches from stage I to stage II, which is in line with its design concept. Model†s are trained with larger learning rates, smaller batch sizes, and fewer epochs, which effectively shortened the training time. Meanwhile, their training processes are stable and robust. There is no gradient explosion even when the step jump occurs, which confirms that SS classification and DSM regression are indeed goal-aligned, and the joint learning of the two is reasonable. There is another point that also confirms the above view (refer to Figs. 5, 6, 9, and 10). The fourth to sixth rows in the figures are the test results of DANet†, PSPNet†, and U-net†, respectively. Comparing the DSM reconstruction and SS visualized results pairwise, it can be seen that even though the height prediction quality of individual pixels varies between methods, the general outlines of semantic categories (or the structure of the ground object distribution) are similar. In fact, the visualization characteristics of semantic labels and DSM are consistent, as mentioned before, only the former is discretized and the latter is continuous. Furthermore, the raw U-net, PSPNet, and DANet are also performed SS tests on Vaihingen and Potsdam, and the results are presented in Tables III and IV, respectively. Test results explicitly prove that the joint multitask framework also helps improve SS accuracy. All in all, the proposed joint multitask framework along with the joint SD loss not only effectively achieves simple and robust training for DSM regression task, but also the advantage of improving the accuracy of SS.

2) Role of the Two-Stage Strategy of Joint SD Loss: This section investigates the role of the two-stage strategy of joint SD loss on model training and performance. Srivastava *et al.* [16], Carvalho *et al.* [17], and Zheng *et al.* [23] take the weighted sum of classification loss and regression loss as the training objective, just like the object detection task does. The baseline method also adopts this manner to conduct ablation study, which is denoted as baseline‡. Specifically, baseline‡ is trained on Vaihingen with (3) as the loss function for 150 epochs (about 100 000 iterations)

$$L = e^{L_{\text{cls}}} + L_{\text{reg}} \tag{3}$$

note that (3) does not use the conventional weighted sum but adopts the form of stage II of joint SD loss, in order to make the baseline and baseline‡ a close comparison. Moreover, the weighting and exponent are both to keep L_{cls} in the same order of magnitude as L_{reg} without being overwhelmed, which are logically consistent. The initial learning rate is set to 0.001, and other details are the same as in the baseline. The training loss curves of the baseline and baseline‡ are plotted in Fig. 12 for comparison, which clearly reflects that the two-stage strategy of joint SD loss can make the model converge faster and more robustly than the parallel cooptimization of classification and regression tasks. In terms of model performance, the quantitative experimental results are shown in Tables I and III. The baseline‡ has trained for 2.5 times as many iterations as the baseline, but its DSM reconstruction accuracy as well as SS accuracy are both inferior to the baseline. The baseline‡ should

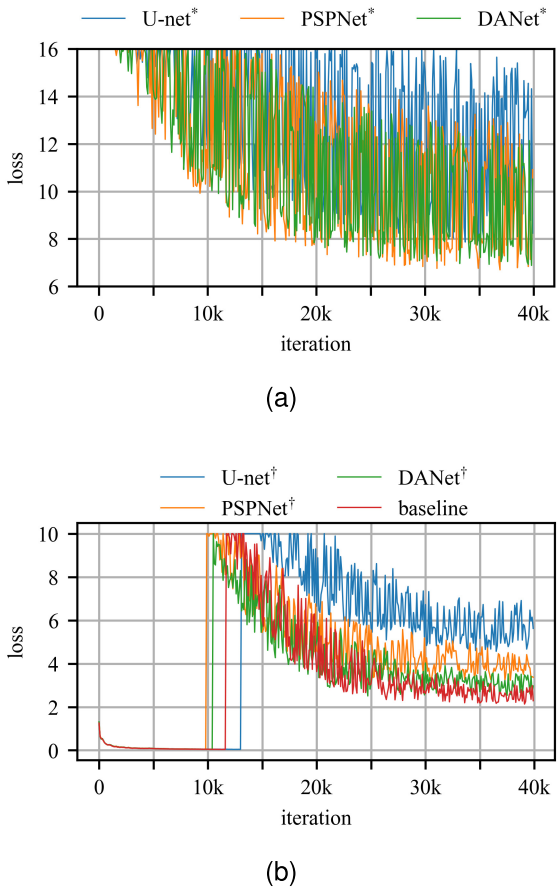


Fig. 11. Loss values of (a) model*s and (b) model†s (including the baseline) trained on Vaihingen for DSM reconstruction.

be able to achieve better performance while being inevitably time-consuming and laborious. The above experimental results can be explained as follows: the weighted sum of classification and regression losses is a common and effective fashion of loss function in object detection, while only a small number of positive samples (usually a few dozen at most) participate in the calculation of regression loss in object detection [35]. In particular, if the object direction is not considered, the number of regression predictions is 4× the number of positive samples (i.e., the positions and sizes of the bounding boxes), which is generally a few hundred values at most. However, in DSM reconstruction, all the locations of the final feature map participate in the calculation of regression loss, and the number of regression predictions is very large. Taking the input size of this article as an example, there are 128 × 128 (size of the final feature map of the regression branch, before being upsampled by a factor of 4 to become the output), that is, more than 16 000 values to be regressed. In this case, having such a complicated regression task involved from the very beginning makes the model quite difficult to train, as shown in Fig. 12, the curve of baseline‡ oscillates significantly compared to those of the baseline. The baseline‡ uses a smaller learning rate (one-tenth of the baseline) in order to keep the model from divergent, so it converges slowly; at the same time, the hyperparameters have to be carefully tuned during training. In addition, the classification capability of the baseline‡ has

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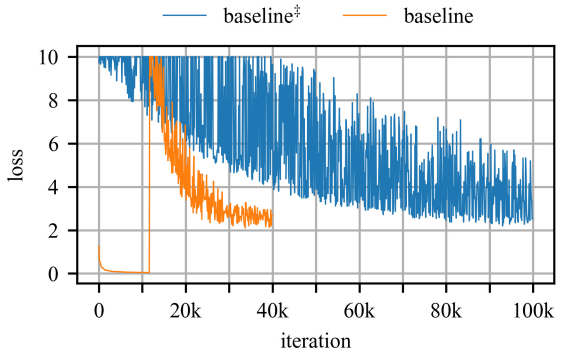


Fig. 12. Loss values of the baseline and baseline‡ trained on Vaihingen for DSM reconstruction.

also been hindered by the unstable regression task, resulting in its longer training but poorer SS performance than the baseline, as shown in Table III. In contrast, the two-stage strategy of joint SD loss allows the "pretrained" classification model to perform regression optimization in a "semi-ideal" state and constrains the floating range of continuous predictions within semantic discrete intervals. Thus, the difficulty of complicated large-scale regression tasks is greatly alleviated. All in all, the two-stage strategy of joint SD loss is more applicable and effective for the specific semantic joint DSM regression task.

3) Role of the FFM Backbone and the FIU Module: Effects of the proposed FFM backbone and FIU module on task performance are studied in this section, while U-net†, PSPNet†, and DANet† still serve as comparisons. First, Fig. 11(b) reveals that the baseline can achieve a lower loss value than the comparisons in training. It is consistent with the quantitative results in Tables I and II, while the accuracy of DSM reconstructions of the baseline outperforms the comparisons by a large margin. In particular, the best performer of the three in the two groups of experiments is DANet†. On Vaihingen and Potsdam, the RMSEs (mean) of the baseline are 13.7% and 27.9% lower than DANet†, and the ZNCCs (mean) are 3.2% and 2.0% higher than DANet†, respectively. In addition, "*(M)*" in the tables means that the experiment actually adopts multi-GPU distributed training to shorten the time, and the result is converted to the single GPU case. Test data show that the best in terms of efficiency among the comparisons is U-net†, while the baseline model has 71.1% fewer parameters than it, and correspondingly has huge advantages in inference speed and training time. Second, the test results in Tables III and IV verify that the baseline also achieves the best performance on SS tasks. Compared with DANet†, its F1 score and OA improve by 1.66% and 1.58% on Vaihingen, and 3.95% and 1.31% on Potsdam, respectively. The visualized results in Figs. 5, 6, 9, and 10 further intuitively demonstrate the superiority of the baseline on the two types of tasks. In summary, the combination of the FFM backbone and FIU module is quite effective for DSM reconstruction and SS of monocular RS images. On the one hand, this is due to the ameliorated upsampling approach, which is one of the main bottlenecks affecting the decoding accuracy and thus the final task accuracy. On the other hand, this article believes that the lightweight of the baseline model is also an important factor in the accuracy improvement. Does the prediction accuracy of a deep learning model certainly have a positive correlation with its complexity or parameter quantity? A growing number of studies now suggest that this is not necessarily the case. It does not mean that the fitting ability of the model will become worse as the complexity increases, but the difficulty of training will increase. When a complex model is difficult to optimize to a suitable set of parameters, it may not perform as well as a relatively simple model that is easy to train to a competent state. In other words, when complex models are oversaturated for simple tasks, appropriately reducing the number of parameters may have a positive effect, which is what knowledge distillation has been working on.

V. Conclusion

In this article, a semantic joint framework is proposed to deal with the monocular RS image DSM reconstruction issue. The framework extracts complementary features through a shared backbone to improve the performance of the independent task and introduces a two-stage joint SD loss to effectively achieve robustness and convergence of model training. Under the synergy of the elaborated FFM backbone and FIU module, the proposal outperforms or is comparable to SOTA-related works for monocular RS image DSM reconstruction and also ameliorates the inherent limitations of SS in terms of decoding accuracy and efficiency. However, the implementation condition of this method is relatively harsh, that is, DSM and semantic label GTs are both required. How to minimize the input cost, such as resorting to weak supervision, will be the follow-up research direction of this article.

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